MIL-F-43320A(GL)
29 November 1968

SUPERSEDING
MIL-F-43320(GL)
19 March 1965

MILITARY SPECIFICATION

FAN, TUBE AXIAL, FOR VENTILATING MAINTENANCE TENTS

1. SCOPE

* 1.1 This specification covers a tube axial ventilating fan with an electric motor and toggle switch all mounted in a frame assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-580	-	Cord, Flexible, and Wire, Fixture, (Electrical, 0- to 600-Volt Service).
CC-M-636	-	Motor, Alternating-Current, (Fractional Horsepower).
QQ-A-200/8	-	Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, and Structural, Shapes, 6061.
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet.
TT-E-529	-	Enamel, Alkyd, Semigloss.
TT-P-666	-	Primer Coating, Zinc Yellow (Zinc Chromate), for Aluminum Surfaces.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-C-5541	-	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-P-8585	-	Primer, Coating, Zinc Chromate, Low-Moisture Sensitivity.

FSC 4140

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STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Loads 40" x 48" Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

DRAWING

ARMY NATICK LABORATORIES

5-13-2281 - Fan, Tube Axial, for Ventilating Maintenance Tents.

(Miniature copy of Drawing 5-13-2281 identified as figure 1 is attached for information purposes only.)

(Copies of specifications, standards, publications, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, (UL) INC.

Standard No. 62 - Flexible Cord and Fixture Wire

Standard No. 498 - Attachment Plugs and Receptacles

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; Walt Whitman Road, Melville, Long Island, N.Y. 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

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AIR MOVING AND CONDITIONING ASSOCIATION, INC. (AMCA)

AMCA Standard 210-67 - Test Code for Air Moving Devices

(Application for copies should be addressed to the Air Moving and Conditioning Association, Incorporated, 205 West Touhy Avenue, Park Ridge, Illinois 60068.)

3. REQUIREMENTS

* 3.1 Alternate components.- Components offered as equivalent to components specified hereinafter and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal, or equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components. Prior to manufacture of the preproduction sample, the supplier shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirement specified above. The contracting officer, at his own option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of preproduction sample manufacture and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

* 3.2 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production item.

3.3 Standard product.- The fan covered by this specification shall be a regular commercial product of the manufacturer, except for any changes necessary to comply with the specification requirements. All parts shall be new and all fans or components furnished under any one contract shall be identical.

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3.4 Standards and publications.- The cords and wires, plugs, and receptacles furnished with fans shall conform to UL Standards No. 62 and 498, respectively. The fan shall conform to AMCA Standard 210.

* 3.4.1 Compliance.- Prior to approval of the preproduction sample, or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the fans he proposes to furnish under this specification meet the requirements specified in 3.4.

* 3.4.1.1 UL.- Acceptable evidence of meeting the requirements shall be the UL label, a photostatic copy of the UL listing card for the material being offered, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electrical materials and components offered, have been tested and meet the requirements of the applicable UL standard.

* 3.4.1.2 AMCA certification.- Acceptable evidence of meeting the requirements of AMCA Standard 210 shall be the AMCA Certified Rating Seal, or a certified inspection report from an independent testing laboratory indicating that the fan was tested in accordance with the requirements of AMCA Standard 210.

3.5 Materials.- Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.5.1 Tube.- The aluminum tube shall conform to QQ-A-200/8.

3.5.2 Sheet.- The aluminum sheet shall conform to QQ-A-250/11.

3.5.3 Primer.- The primer shall conform to TT-P-666 or MIL-P-8585.

3.5.4 Enamel.- The enamel shall conform to class A of TT-E-529 and shall be olive drab color No. X24087 of Fed. Std. No. 595.

3.5.5 Cord, flexible.- The cord shall be type SO as specified in J-C-580.

3.6 Design and construction.- The fan shall be constructed as specified on Drawing 5-13-2281 and hereinafter. The fan assembly shall consist of a tube axial fan and electrical components mounted in a frame. The tube axial fan shall consist of a motor, housing frame, impeller, and outlet screen. The component parts of the fan assembly shall be connected and wired for immediate use. The completely assembled fan shall have a nominal capacity of 1100 cubic

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feet per minute (CFM) and shall conform to the requirements of AMCA Standard 210 when tested as specified in 4.4.1. The weight of the complete assembly shall not exceed 25 pounds. The capacity shall be rated at standard air (see 6.4).

3.6.1 Motor.- The motor shall be designed to operate at approximately 3450 rpm on nominal 120-volt, single phase, 60-Hertz alternating current, shall conform to CC-M-636, and shall not require the use of capacitors. The motor shall be designed for continuous operation in any position and shall be totally enclosed. The motor windings shall be impregnated or treated to prevent moisture or fungus damage. The motor bearings shall be lubricated with grease and shall have grease fittings unless permanently lubricated bearings are provided.

3.6.2 Housing frame.- The housing frame shall be designed to house the impeller and shall have a thickness of not less than 16 gauge aluminum (0.050 inch nominal). The housing shall allow removal of the fan impeller and motor without removing the housing from the support frame.

3.6.3 Outlet screen.- The outlet screen shall be attached to the discharge flange of the fan housing with means for removal.

3.6.4 Impeller.- The impeller shall be constructed of aluminum and shall be directly connected to the motor and centered in the housing. The impeller shall be statically or dynamically balanced in accordance with commercial practice for operation at rated speed and there shall be no excessive noise or vibration when tested as specified in 4.4.2. When required, all added weights shall be arc or gas welded in place.

3.6.5 Frame assembly.- The frame assembly shall be constructed of aluminum tubing specified in 3.5.1 and aluminum sheet specified in 3.5.2.

3.7 Wiring.- Fixture type splicing connectors shall be used for all wiring connections, except connections for screw type terminals which shall be secured with slotted head pressure terminal connectors.

3.7.1 Motor to switch.- There shall be a type SO flexible cord between the motor and switch. The cord shall allow the tube axial fan to be rotated 90 degrees each side of the vertical position when tested as specified in 4.4.2.

3.8 Operation.- When tested as specified in 4.4.2, the completely assembled fan shall start when the toggle switch is placed on, shall not fail while operating, shall stop when the toggle switch is shut off, and while operating, the current to the motor shall not exceed the data plate rating.

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* 3.9 Electromagnetic compatibility.- When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.3).

* 3.10 Finish.- The fan shall be prepared, primed, and enameled in accordance with the applicable requirements specified hereinafter.

* 3.10.1 Surface preparation.- After all forming, stamping, drilling, and welding, the aluminum components shall be prepared for painting in accordance with class 2 requirements of MIL-C-5541. Surface preparation of rivets is not required.

* 3.10.2 Priming.- Prepared aluminum surfaces, housing, exterior motor surface, switchbox rivets, screws, nuts, and washers shall be coated with primer specified in 3.5.3. Priming shall be accomplished either prior to or after assembly.

* 3.10.3 Enameling.- Primed surfaces shall be coated with enamel specified in 3.5.4. Rivets, screws, nuts, and washers, after priming, shall be enamel coated. The final finish shall be uniform in appearance, free from bare spots, sags, runs, pinholes, blisters, wrinkles, frothing, or other defects characteristic of improper application or cure.

3.11 Marking.-

3.11.1 Identification.- Identification marking shall be on the manufacturer's standard name plate. The plate shall be secured to the fan housing in a manner to be readily viewed.

3.12 Instruction plate.- Each fan assembly shall be equipped with an instruction plate, suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. The instruction plate shall be of the same material as the plate specified in 3.11.1. When an instruction plate is not required, power source shall be indicated on the identification plate specified in 3.11.1.

3.13 Equipment manual.- Manual contents and additional manual distribution shall be as specified (see 6.2).

3.14 Workmanship.- The finished item shall be free of sharp corners or edges.

3.14.1 Aluminum fabrication.- After fabrication, aluminum plates, strips, and tube shall have no visible twists or kinks. Fabrication shall produce square corners and bends of uniform size and shape.

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3.14.2 Welding.- The surface of parts to be welded shall be free from scale, grease, and other foreign matter. Welds shall be free from pits, holes, or fissures. The deposited metal shall be fused with the base metal along all surfaces and edges of union and shall penetrate the full depth of the joint.

3.14.3 Fastenings.- Bolts shall have at least 1-1/2 threads exposed past the nut and the nut shall be tight. Rivet heads shall be concentric with the hole and in contact with the surface of the metal. Washers or lock washers shall be provided where necessary. Cotter pins shall have end edges broken and shall be bent back against metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined in accordance with table II, specified dimensions and tested in accordance with 4.4.1 through 4.4.3.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 Process inspection.- Inspection shall be made during the manufacturing process for the requirements specified in table I to establish that no deviation is made from indicated requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

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TABLE I.- Process inspection

Requirement	Requirement paragraph
Surface preparation	3.10.1
Application of primer	3.10.2

4.3.3 End item inspection.- The lot shall be the number of fans offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated fan assembly.

4.3.3.1 Visual examination.- Examination of the fan assembly shall be in accordance with the classification of defects in table II. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified		X
	Not finished where required	X	
Construction and workmanship (applicable to all components and assemblies)	Part missing or not in proper alignment	X	
	Sharp corners or edges	X	
	Not connected or joined as specified	X	
Welding and brazing	Burn hole, cracked, fractured, or otherwise not fused	X	
	Scale, grease, or foreign matter		X
	Pits, holes, or fissures	X	
Threaded fasteners	Loose		X
	Not at least 1-1/2 threads beyond the nut (when applicable)		X
Cotter pins	Edges not broken, or not bent back against metal		X

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Rivets	Head not in contact with the metal or not concentric with the hole		X
Wiring	Not connected as specified	X	
Marking for identification	Missing, incomplete, not legible	X	
Equipment manual	Missing, incomplete, not legible	X	
Instruction plate (when required)	Missing, incomplete, not legible Not type specified	X	X

4.3.3.2 Dimensional examination.- An examination shall be made of fan assemblies for compliance with specified dimensions. Any dimension that is not within the specified tolerances shall be classified a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item.- When a preproduction sample is not required, the initial production unit shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3, when applicable. Any nonconformance shall be cause for rejection.

* 4.3.4 Codes and standards compliance.- The supplier shall furnish the contracting officer or his authorized representative with proof of compliance with the code and standards requirements of 3.4.1.1 and 3.4.1.2.

* 4.3.5 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot shall be the number of shipping containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

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TABLE III.- Examination of preparation for delivery

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Bulged or distorted container.

4.3.5.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be made in accordance with that specification.

4.4 Tests.-

4.4.1 Fan performance.- The fan shall be connected to power and operationally tested to determine compliance with 3.6. The test shall be in accordance with AMCA Standard 210. All results shall be corrected to standard air conditions (see 6.4).

4.4.2 Operation.- Connect the fan to 120-volt, single phase, 60-Hertz power and operate and rotate 90 degrees to determine compliance with 3.7.1 and 3.8. While operating, the amperage to the motor shall be determined for compliance with 3.8 and the noise and vibration shall be checked for compliance with 3.6.4.

* 4.4.3 Electromagnetic compatibility test.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with test methods CE03, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer, for approval, the Interference Control Plan, the EMI/EMC Test Plan, and the Electromagnetic Test Report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production, as applicable, shall be used as a model for all production units.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- The tube axial fan shall be secured within the frame in such a manner as to prevent movement while in transit. The power cable shall be completely coiled and secured to the frame. Technical publications shall be placed in a paper envelope.

5.1.2 Level C.- Fans shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Each fan, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Technical publications shall be placed in an unused portion of the shipping container. Fiberboard inserts shall be used to provide clearance between projecting parts of the fan and the interior surfaces of the shipping container, and to immobilize the fan within the container. Closure, waterproofing and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the container specification.

* 5.2.2 Level B.- Each fan, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic of PPP-B-636. Technical publications shall be placed in an unused portion of the shipping container. Fiberboard inserts shall be used to provide clearance between projecting parts of the fan and the interior surfaces of the shipping container, and to immobilize the fan within the container. Closure shall be in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified, (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification. The box material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Fans, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

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* 5.3 Palletization.- Unless otherwise specified (see 6.2), fans packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the container patterns specified in MIL-STD-147, the container pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The tube axial fan covered by this specification is intended to be used primarily to ventilate maintenance tents.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article.- When preproduction sample is not required (see 3.2).
(c) When electromagnetic compatibility is required (see 3.9).
(d) Manual contents and distribution of additional manuals (see 3.13).
(e) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant box shall be used in level B shipment (see 5.2.2.1).
(g) When palletizing is not required (see 5.3).

* 6.3 Electromagnetic compatibility.- When electromagnetic compatibility is required, the contracting officer should submit a copy of the Interference Control Plan, the EMI/EMC Test Plan, and the Test Report required (see 4.4.3) to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey, 07703, for approval.

* 6.4 Standard air.- All reference made in this specification to air pressure, volumes, and velocities shall be considered as referring to standard air. Standard air shall be interpreted as dry air at a temperature of 70 F., a barometric pressure of 29.92 inches of mercury and a density of 0.075 pounds per cubic foot.

* 6.5 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions)

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Custodian:

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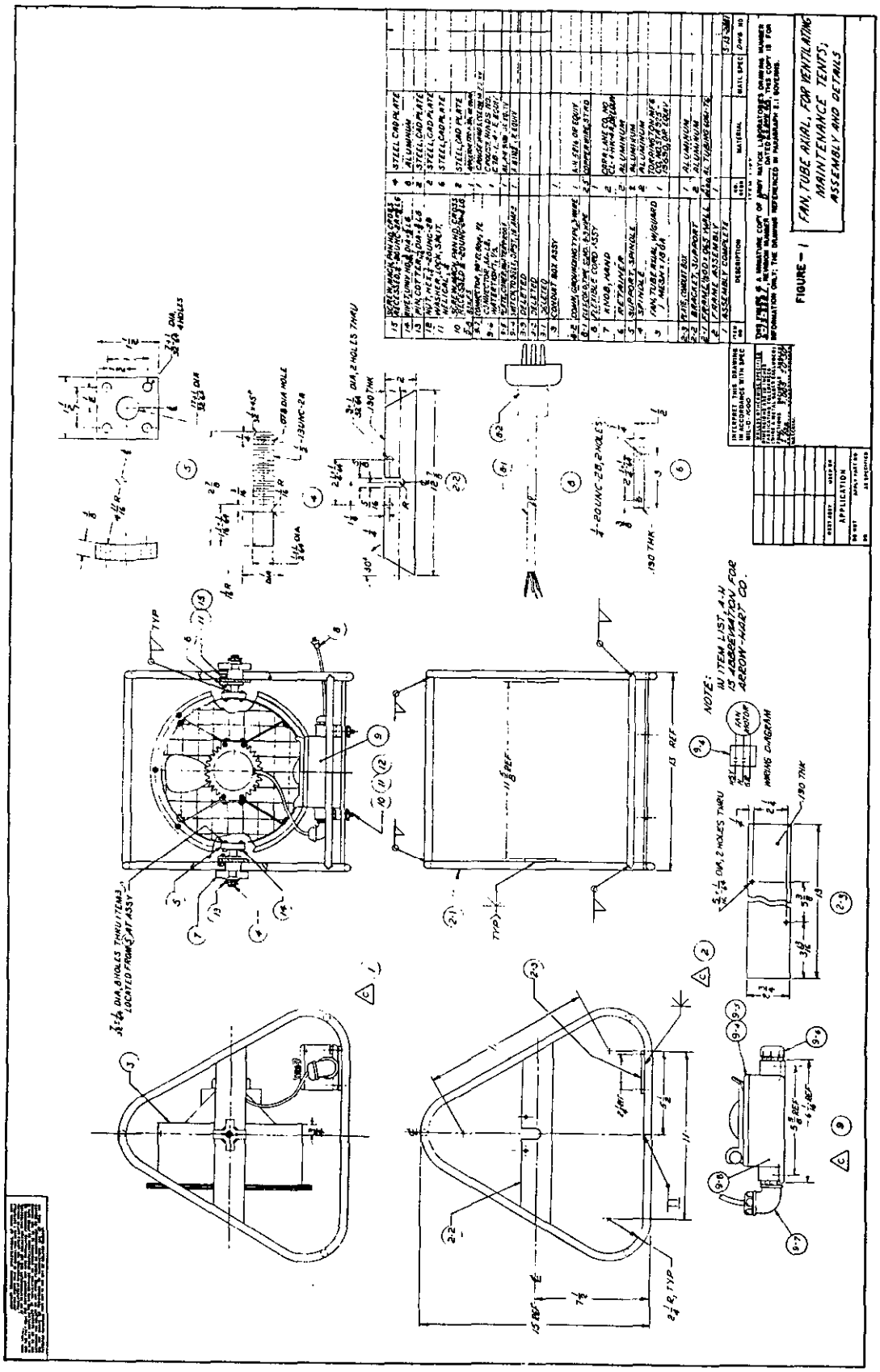


FIGURE - 1 FAN, TUBE AXIAL, FOR VENTILATING MAINTENANCE TENTS; ASSEMBLY AND DETAILS

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